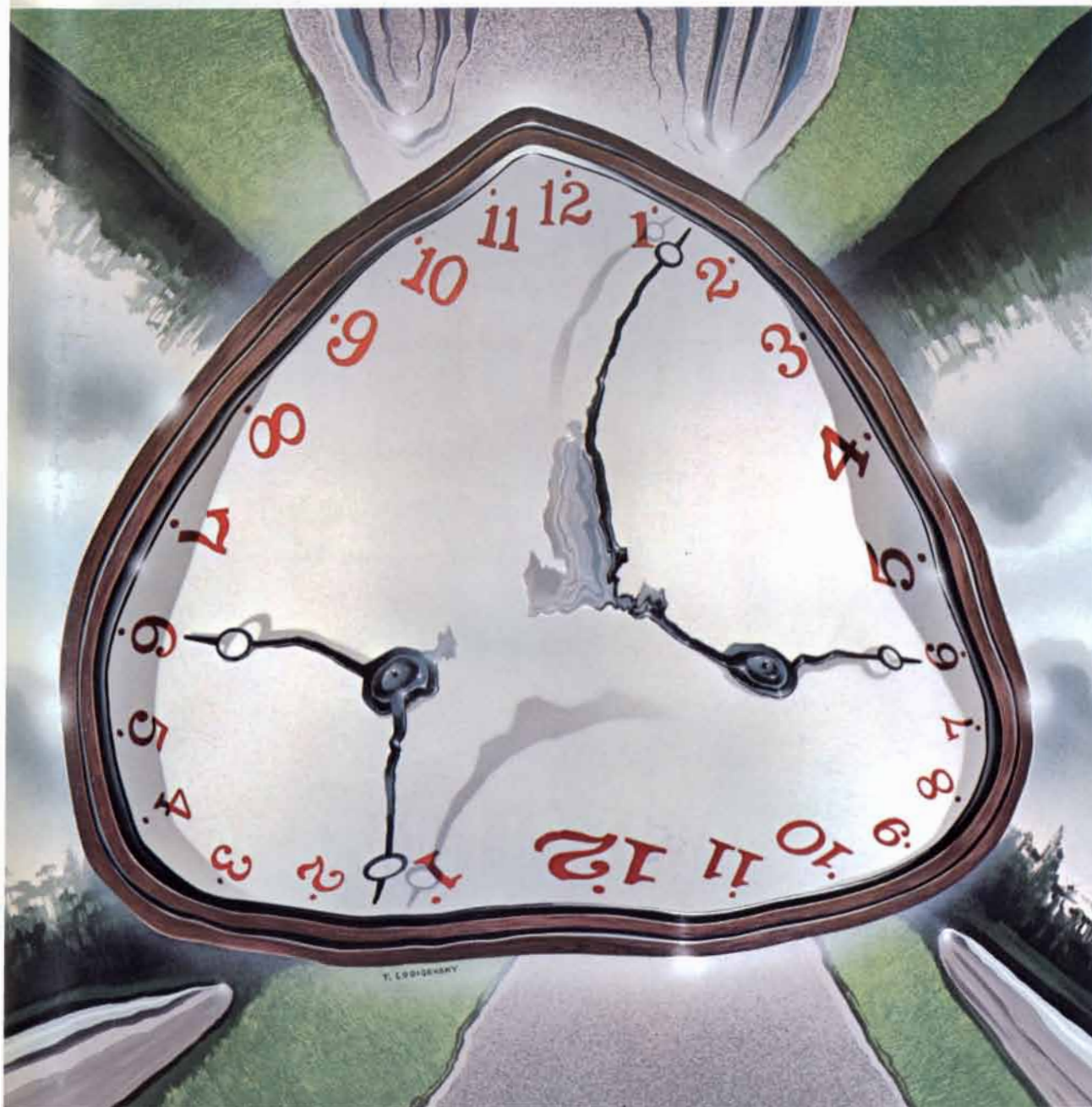


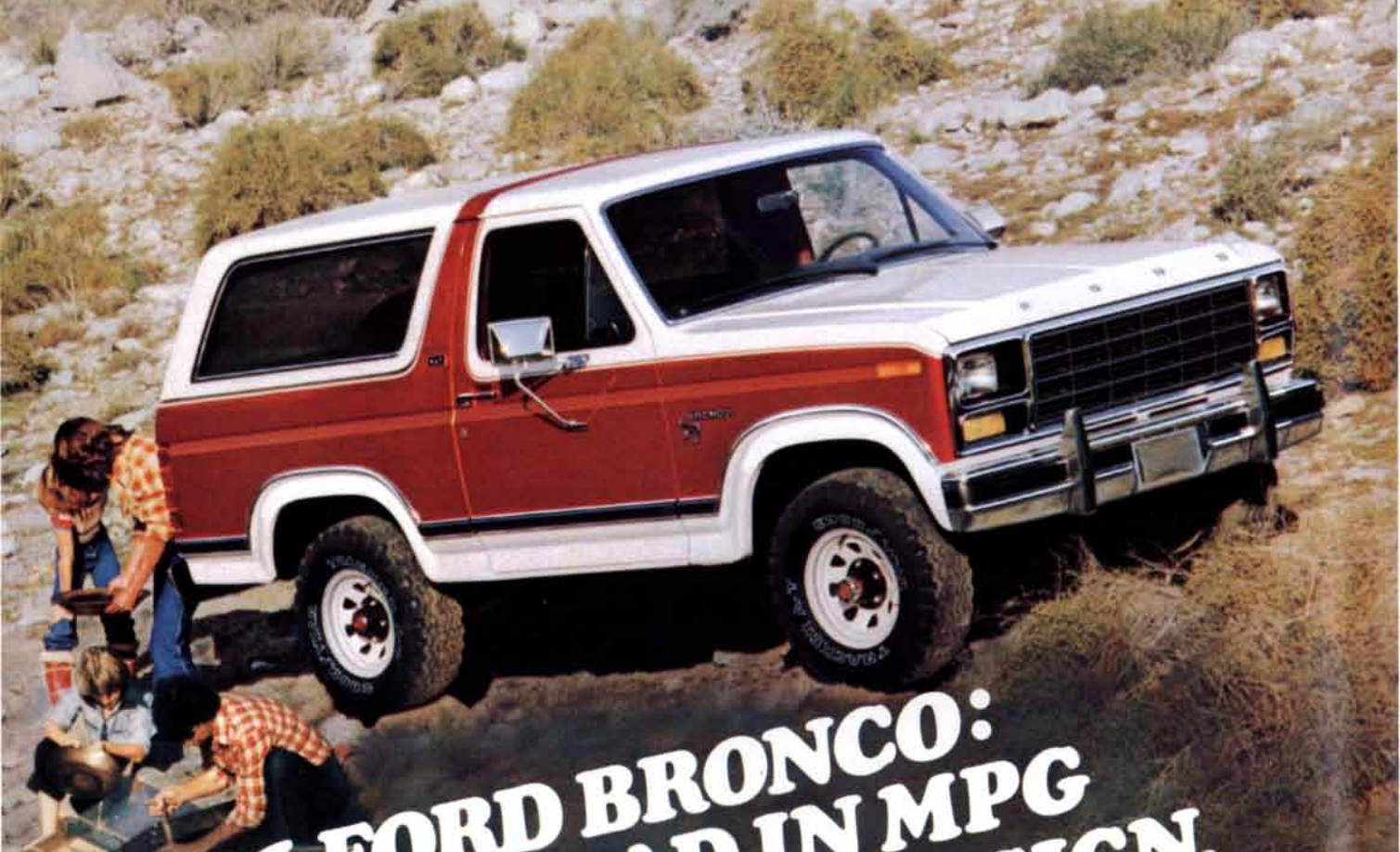
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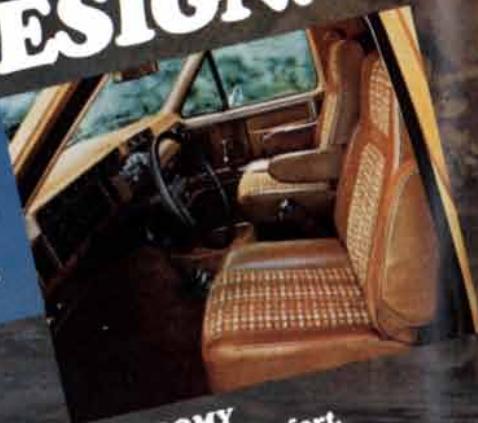
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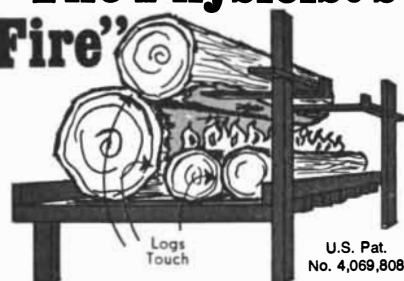
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THE COVER

The painting on the cover shows the image of a clock face reflected in a mirror with three lobes, or hills, and three valleys. A single, continuous image is formed but the clock face is doubled, so that it spans a total of 24 hours. If the image were observed for 12 hours, while the hour hand of the real clock made a full revolution, each of the two hour hands in the image would make just half a revolution. Only after 24 hours would each hour hand complete a full turn. The reflecting properties of various surfaces can be analyzed in terms of a few basic curved mirrors and the boundaries that join them (see "Mirror Images," by David Emil Thomas, page 206). The forms around the clock face are a distorted image of the artist's lawn, where the mirror and the clock were set up.

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THE AMATEUR SCIENTIST

The physics of the patterns of frost on a window, plus an easy-to-read sundial

by Jearl Walker

One of the delights of winter for me is the frost that decorates my windows. The crystals appear in a variety of shapes and designs. Some are long and rodlike; others mimic flowers. Each crystal has developed in the supercooled water condensed from the air near the window. Some unseen nucleating agent, such as a dust particle, initiated the process, starting with a single crystal and then spreading to cover an entire windowpane.

Which of the several possible shapes appears depends primarily on the temperature in the vicinity of the crystal, but the relation is not fully understood. Even a difference in temperature of a few degrees can be critical in determining the shape. Since several different forms can be seen growing on the same window, usually at different heights, the temperature must vary by at least a few degrees over the window. The frost photograph on the opposite page was sent to me by Gera Dillon, a professional photographer in Morin Heights, Quebec, who delights in the frost on his windows.

The basic ice-crystal structure is hexagonal. The plane of the hexagon is called the basal plane and the axis perpendicular to it the c axis. Three axes labeled the a axes pass through the sides of the hexagon. The ice on a window is birefringent when it is viewed along any direction that is not parallel to the c axis. The term refers to the fact that two different values for the index of refraction can affect light as it passes through the ice.

I explained the general properties of birefringent materials in this department for December, 1977. To summarize briefly, a birefringent material has a "fast" axis and a "slow" one. The index of refraction depends on how the light is polarized when it passes through the material. The index is higher if the light is polarized parallel to the slow axis and lower if it is polarized parallel to the fast axis.

Suppose linearly polarized light (light polarized along a single axis perpendicular to the ray) is directed through the

birefringent material with its axis of polarization at an angle to both the slow and the fast axes of the material. The polarization can be separated mathematically into two components, one parallel to the slow axis and one to the fast. These two waves were in phase when they entered the material, but because of the different indexes of refraction they probably emerge with a different phase relation.

The result is that the emerging light probably has a polarization different from that of the incident light. The new polarization might still be linear but with the axis of polarization oriented differently. The emerging light could also be circularly or elliptically polarized, which means that the axis of polarization rotates about the light ray as the light passes.

If the emerging light encounters a polarizing filter, it may or may not pass through the filter, depending on the relative orientation of the polarization axis of the light and the polarization axis of the filter. Birefringent materials are often analyzed by being placed between two polarizing filters. Light passing through the first filter becomes linearly polarized. It then passes through the birefringent material and probably is changed in polarization. When it encounters the second filter, it is transmitted or absorbed according to the relative polarization of the light and the filter.

The change in the polarization of the light by the birefringent material depends on the indexes of refraction for the slow and fast axes, the thickness of the material and the wavelength of the light. If white light is sent through the two filters and the birefringent material, some wavelengths will emerge from the material polarized in such a way that they can pass through the second filter and be seen. Other wavelengths will not pass through. Although white light is directed into the first filter, only certain colors emerge from the second. If either of the filters or the birefringent material is rotated, the colors that emerge from the second filter shift.

Frost on a window can be analyzed in

much the same way because ice is slightly birefringent. In order to see the colors put a polarizing sheet on each side of the frost-covered window. All the crystals with the proper thickness and orientation will contribute colors. The crystals with c axes parallel to the line of sight will not contribute, however, because such an orientation eliminates birefringence.

The colors from the frost can be seen without polarizing filters if some of the frost has melted into a pool at the base of the window. The light scattered from the sky can be strongly polarized by mechanisms I described here in January, 1978. If the window is illuminated with such light, the first filter is not needed. If the light passes through the frost and then reflects from the pool of water, the reflection can act as the second filter, because a reflection can polarize light. When you look into the pool, you see colored versions of the colorless frost patterns on the window above the pool.

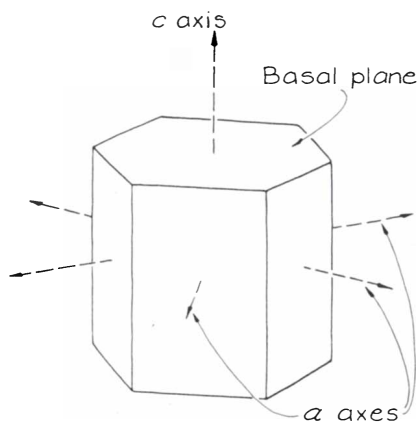
C. K. Sloan of Arapahoe, N.C., has designed an unusual type of sundial. Unlike many other types, his "analemmic sundial" is corrected for the effects of the declination of the sun and the latitude and longitude of the sundial. It can be read either in universal time or in the local apparent time ordinarily indicated by a sundial.

Sloan's sundial is a simple one: a vertical post casts a shadow onto a grid he has devised. In some versions of the sundial the grid is several inches across; in larger versions it measures as much as 75 by 150 feet. In any version the grid is beautiful, reminding me of ancient representations of the passage of the sun through the sky.

Many ordinary sundials employ a gnomon, or shadow-casting object, pointing obliquely at the celestial pole, the zenith of the sky over the geographical pole. The shadow is cast on a grid of lines extending from the base of the gnomon. When the sun is at its highest point of the day, the shadow falls directly below the gnomon on the noon mark, one of the radial lines extending from the base.

Although this garden variety of sundial is attractive, telling time with it calls for conversion tables and a little work. The dial is geared to the passage of the sun, not to a clockwork. The time read directly on a sundial is the local apparent time. When the sun is at its highest point, the local apparent time is noon. When the sun passes through an arc of 7.5 degrees in the sky, a half hour has elapsed.

The system is straightforward, but it presents at least two problems. One of them is that the local apparent time depends directly on the longitude of the sundial; sundials at different longitudes will differ in time. The other is that the length of an hour of local apparent time



The structure of an ice crystal

changes in the course of the year because of the shape of the earth's orbit around the sun. As a result the apparent speed of the sun across the sky differs with the passage of the months. Each hour of the local apparent time remains proportional to the sun's movement through an arc of 15 degrees, and so the length of an hour changes.

Another system of time, local mean time, is similar to local apparent time except that it is geared to the passage of an imaginary sun moving uniformly across the sky throughout the year. The uniform motion means that the length of an hour does not change with the passage of the months. The difference between the two systems of time is called the equation of time. Tables of the difference can be found in reference books devoted to the accurate timing of the stars and the planets.

During two periods of the year the sundial is said to be slow because its time lags behind the local mean time. During the other two periods of the year the sundial is fast. To convert the time

given by a sundial in the slow period into the local mean time add the equation of time for that particular day to the local apparent time given by the sundial. In the fast periods the equation of time is subtracted from the local apparent time. This procedure requires, of course, ready access to a table of the equation of time. Moreover, the observer is still confronted with the problem that the local mean time is only local.

The time on a clock is usually "standard" or "daylight-saving" time. As is well known, the world is divided into time zones where regardless of the actual longitude of the clocks each clock keeps the local mean time for a certain meridian running through the zone. This special meridian is called the standard meridian for its time zone. The advantage of the system is that the time is the same for all clocks in a particular zone.

Converting the time read directly on a sundial (the local apparent time) into the time read on a clock (the standard time) calls for two operations. First the sundial's time must be converted into local mean time by using a table listing values for the equation of time. Then the local mean time for the sundial's location must be converted into the standard time for its time zone.

This second conversion amounts to determining the difference in longitude between the sundial and the standard meridian for the zone. If the sundial is west of the standard meridian, its local mean time will be late. The clock indicates noon when the imaginary sun passes over the standard meridian. Only later will the imaginary sun pass over the sundial. Hence the sundial's local mean time lags behind that of the clock. If the sundial is east of the standard meridian, its local mean time is ahead of the clock's. The greater the difference in longitude between the standard meridian and the sundial, the greater the dis-

crepancy (four minutes of time for each degree of difference in longitude).

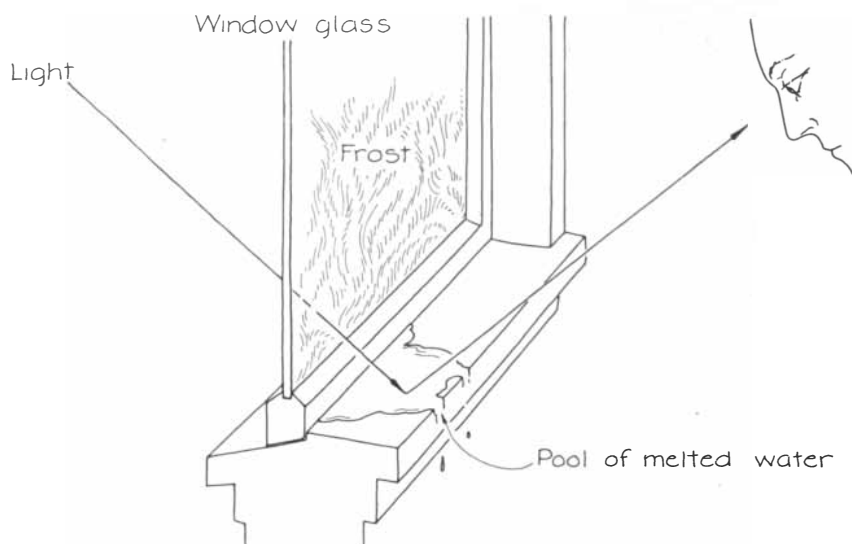
Attempting to read clock time from a common sundial can therefore be a tedious procedure. Sloan's design enables the observer to read the clock time directly from the sundial with no need for tables or further calculations. All the corrections are built into the design. Instead of having an oblique gnomon and following the rotation of the shadow around its base, Sloan has a vertical gnomon and follows the tip of the shadow. In the course of the day the tip of the shadow moves across the grid laid out around the post. Corrections for both the equation of time and the distance of the sundial from the standard meridian in its time zone are incorporated into the design of the grid.

The path of the tip of a vertical gnomon's shadow on a horizontal surface normally does not go directly from west to east as the sun passes through the sky. To determine the path Sloan erected a vertical gnomon and marked two key points near it. Due south of the gnomon, at a distance found by multiplying the height of the gnomon by the cotangent of the sundial's latitude, he marked what he called the ecliptic point. It is from this point that the radial lines, one line for each half hour of the day, were to be extended.

Sloan next determined another point (marked *N* in the top illustration on the opposite page) that was due north of the gnomon and at a distance consisting of the gnomon's height multiplied by the tangent of the latitude. Through this point he drew a straight line, called the equinoctial line, that ran east and west. The line is special because on the two days of equinox (March 21 and September 23) the tip of the shadow from the post travels along it. At noon on those days, when the sun is directly overhead, the shadow's tip falls at *N*.

In 30 minutes the earth turns 7.5 degrees, and the tip of the shadow moves along the equinoctial line by a distance equal to the tangent of 7.5 degrees multiplied by the secant of the sundial's latitude and by the height of the gnomon. Sloan marked off units of this distance on the line for half-hour intervals up to five hours before and after the noon point. He next drew radial lines extending from the ecliptic point (the point south of the post) to the half-hour marks. I shall call these lines the half-hour radial lines. Sloan could then read the time from his sundial for two days of the year: the two days of the equinox. On those special days the tip of the shadow first touched the west end of the equinoctial line and then traveled along the line, passing over the half-hour marks until it reached the noon point. Thereafter it continued along the line to the east.

I had originally thought the shadow would move in the same way on the rest



How to see color in a frost pattern

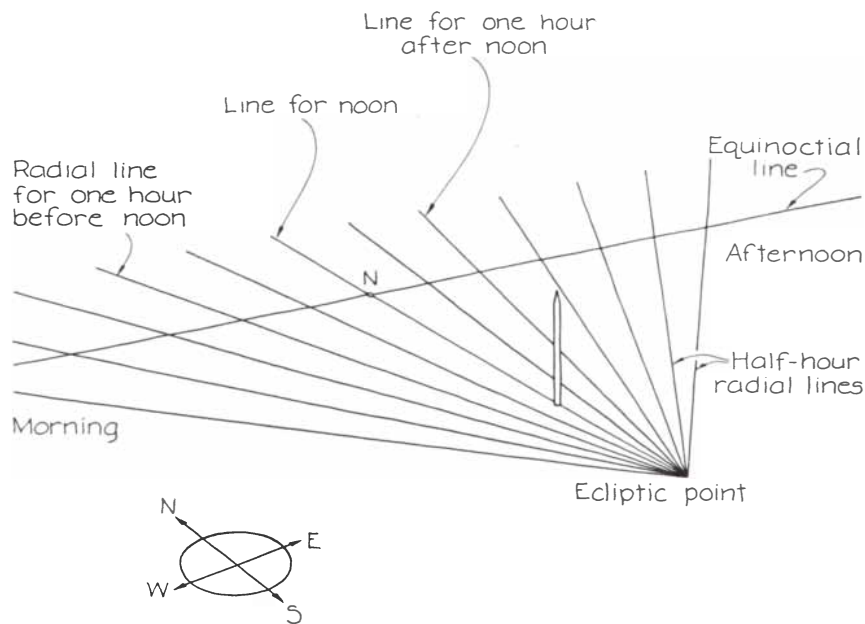
of the days in a year, but it does not. At other times the tip of the shadow falls north or south of the equinoctial line, depending on the season. More precisely, its distance from the line depends on the maximum height of the sun during the day. That height, which is commonly called the sun's declination, is taken to be zero on the equinoxes. In the fall and winter the declination is a negative number because the maximum height of the sun in the sky is relatively low. In the spring and summer the declination is positive because the maximum height is relatively high.

At each equinox the tip of the shadow falls at *N* on the equinoctial line when the sun reaches its highest point in the sky. In the fall and winter, since the declination is less than zero, the sun passes through its noon point lower in the sky and the shadow's tip falls north of the equinoctial line. In the spring and summer the shadow falls south of the equinoctial line and closer to the gnomon.

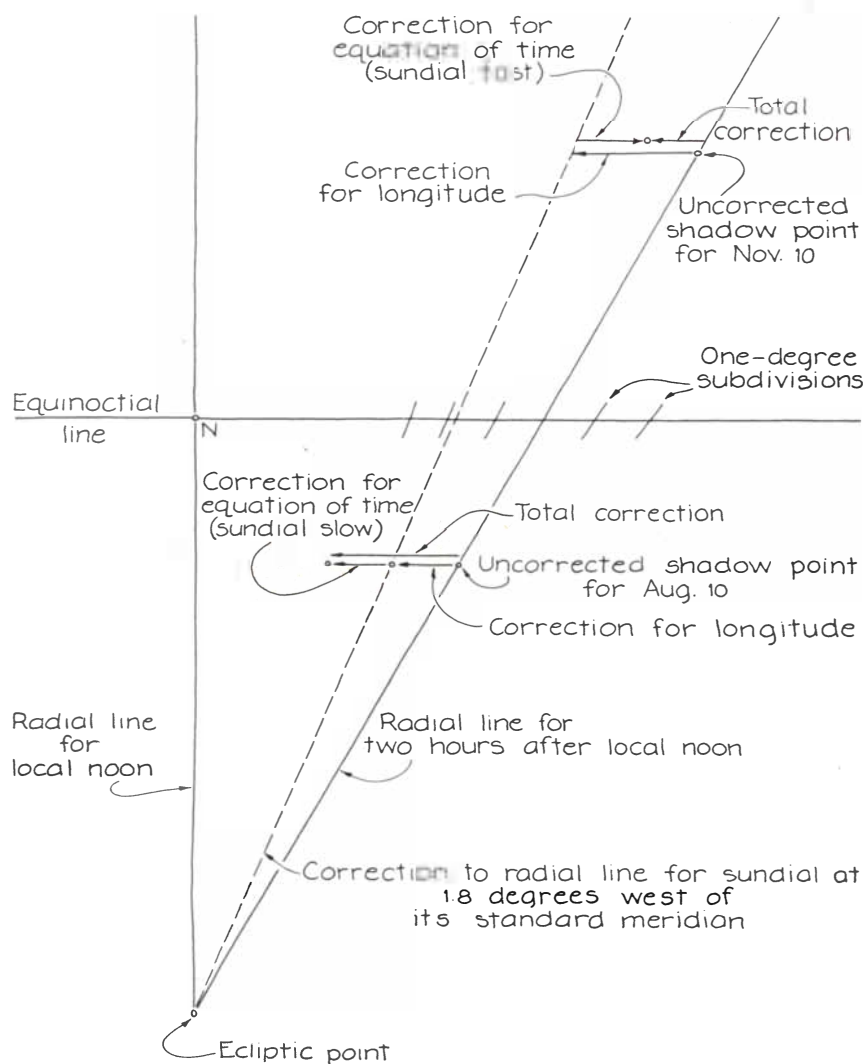
The calculations for the noon positions of the shadow's tip are easy. The distance from the gnomon is found by multiplying the height of the gnomon by the tangent of an angle equal to the algebraic subtraction of the declination of the sun from the latitude. In the spring and summer a positive value for the declination is substituted in the expression and then subtracted from the latitude. In the fall and winter a negative value is substituted, which means that the absolute value of the declination is actually added to the latitude because of a cancellation of negative signs.

The noon points for the shadow are easily determined in this way, but the half-hour points for the rest of the day are not. On an equinox the half-hour points fall on the equinoctial line, which is straight. On any other day of the year the shadow follows a curved path. In the fall and winter it begins the day toward the northwest, moves along a curved line toward the southeast until it reaches the noon point for that day and then continues toward the northeast along a curved line symmetrical with its morning path. The curved path in the spring and summer begins toward the southwest, moves to the northeast to reach the noon point and then curves to the southeast.

Sloan derived equations with which he could calculate the position of the shadow's tip for any time of the day and for any day of the year. With the results he could mark a grid around the post to indicate the local apparent time. I wrote a computer program to do the calculations for a sundial in the Northern Hemisphere. It is easily modified for one in the Southern Hemisphere. The program, which is shown in the illustration on page 238, is written in the computer language called Level II Basic for a Radio Shack TRS-80 home computer. (If you want a copy of the original equa-



A basic grid for a sundial



Corrections of shadow-point locations

tions, send a stamped, self-addressed envelope to me at the Physics Department, Cleveland State University, Cleveland, Ohio 44115. A table of the results of the computation for your latitude can be obtained from Sloan at Route 2, Box 236, Arapahoe, N.C. 28510. In addition he would like to hear how your sundial works out.)

The program calculates the locations of the shadow points along the half-hour radial lines that extend from the ecliptic point. Line 10 contains the latitude of the sundial, which I took as 35 degrees north, and the height of the gnomon, which I took as one meter. (The quantity "RPD" converts degrees into radians, which is the unit required by the trigonometric functions farther along in the program.) When line 20 is reached, the computer asks me to enter a date and the sun's declination (in degrees) for that date. Lines 80 through 120 calculate the locations of the shadow points for 10 times before noon and for noon itself. Since the results will be the same for the equivalent times after noon, the locations are actually computed for 21 times of the day. (Some of the results will seem erroneous if you forget that in winter the sun rises late and sets early.)

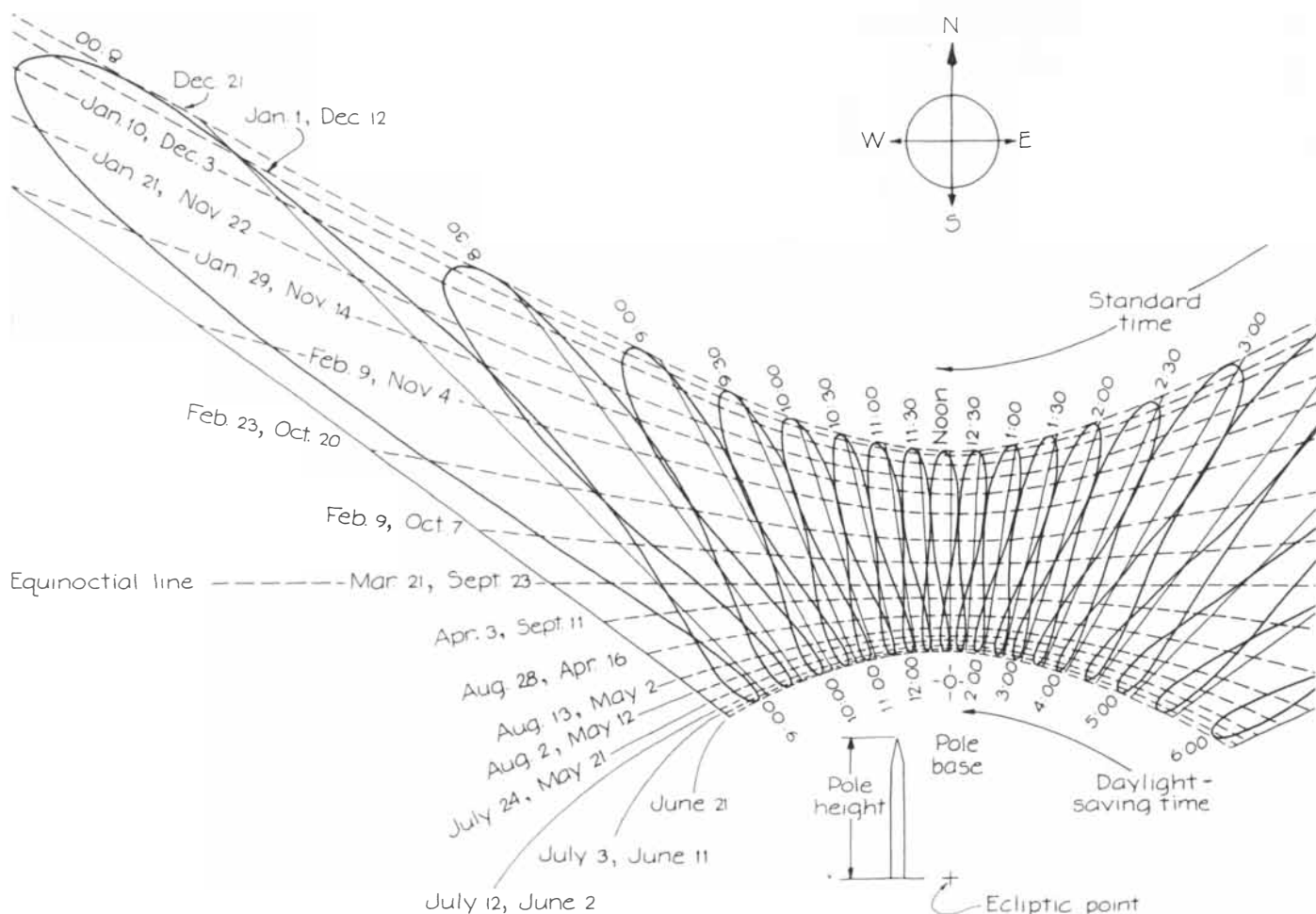
Line 130 displays on the computer screen the time (in hours with respect to noon) and the distance (in meters) for the shadow-point locations. For convenience in plotting the results the distance is given with respect not to the ecliptic point but to the equinoctial line. For example, suppose that at a time of two hours before noon, that is, for a local apparent time of 10:00 A.M., the distance printed on the screen is .2 meter. I would mark the result on the radial line corresponding to 10:00 local apparent time. The point would be at a distance of .2 meter from the equinoctial line on the north side. A similar point would be marked on the radial line corresponding to two hours after noon, since the morning and afternoon patterns are symmetrical. If the distance indicated by the computer is negative, the point is to be marked on the south side of the equinoctial line.

I run the program to cover 34 days spaced approximately evenly through the year. Included are the days of the equinox (when the declination is zero) and the winter and summer solstices (when the declination is extreme). Following Sloan's instructions, I choose the days to achieve a total of 17 different

declinations. Thus each date has a partner on which the sun has the same declination. For example, November 14 and January 29 both have associated declinations of approximately -18 degrees.

Plotting these results for the locations of the shadow points would in itself prove little. For any particular time of the day, say 2:00 P.M. local apparent time, I would plot the distances on the corresponding radial line extending from the ecliptic point. The position of the shadow point for June 21 would be closer to the ecliptic point than that for December 21. The other points would be plotted somewhere between these two extremes. Still, they would all be on the same half-hour radial line from the ecliptic. The resulting pattern would be similar to that of a common sundial.

The power of Sloan's sundial is evident when these shadow-point locations are corrected in two ways. First, for the conversion to local mean time they are corrected by the equation of time. Then for the conversion to standard time they are corrected for the difference in longitude between the sundial and the local standard meridian. Both corrections amount to a rotation of the shadow points around the ecliptic point.

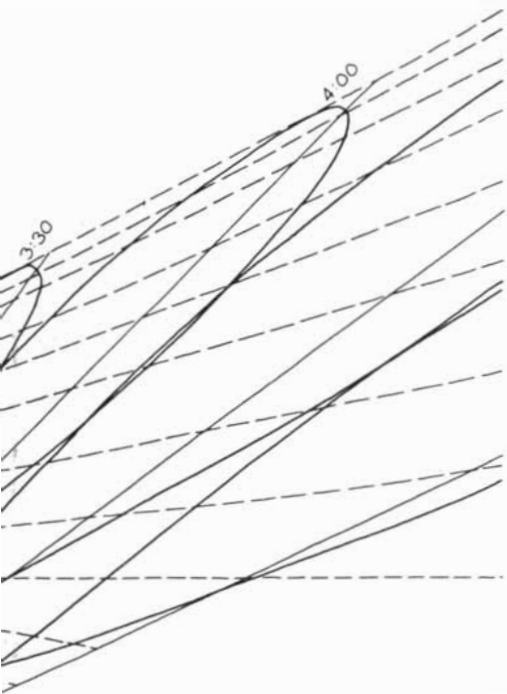


C. K. Sloan's pattern for an "analemmic sundial"

The program computes the total rotation to make the two corrections.

The second correction is a simple one. If the sundial is to the west of the standard meridian, its local mean time is behind the local mean time on the meridian; if it is to the east, its local mean time is ahead. The adjustment for the longitude is made in line 10 of the program, where "DL," the difference in longitude, is entered. I have imagined a sundial 1.8 degrees west of the standard meridian. For a sundial situated elsewhere the number must be changed. If the sundial is west of the standard meridian, the longitude difference is entered as a positive number. If it is to the east, the difference is a negative number.

The correction for longitude is easy because it is the same correction for all the shadow points on the grid. In my example for a sundial 1.8 degrees west of its standard meridian all the shadow points on the grid would be rotated about the ecliptic point by an angle of 1.8 degrees to the west. The appearance of the grid would be unaltered, since all the shadow points for any particular time would still lie in a straight line extending through the ecliptic. The only difference after this correction is that



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My name is Dr. George McCarty. I teach math at the University of California. I wrote this guidebook to cut through the confusion. It does just that — with worked-out examples, simple exercises and practical problems — all designed to work on your calculator!

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If you're in the biological, social or physical sciences, you'll be doing Bessel functions, carbon dating, Gompertz' growth curves, half-life, future value, marginal costs, motion, cooling, probability, pressure — and plenty more (even differential equations).

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for RPN Calculators') writes:

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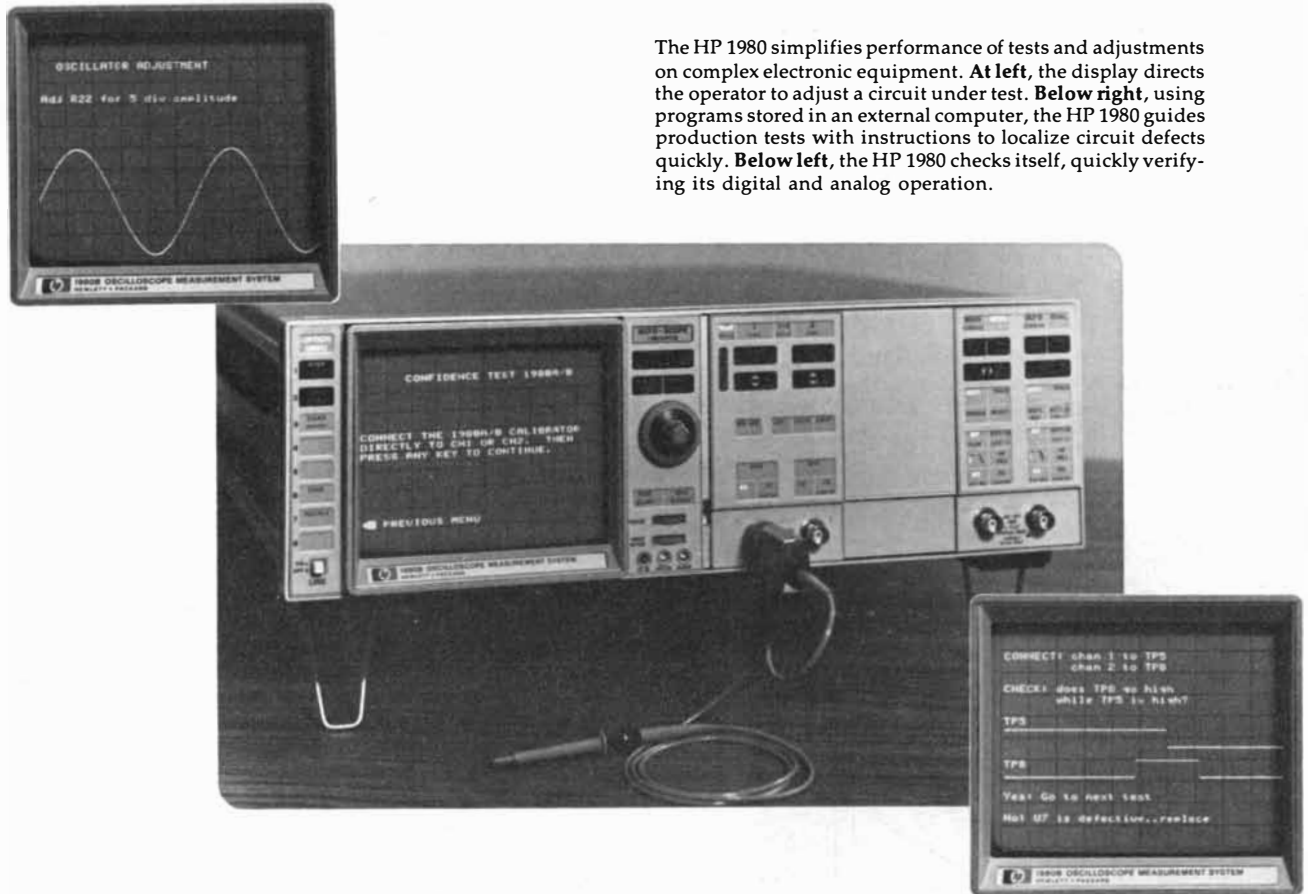
Professor Glyn Daniel, Faculty of Archaeology and Anthropology in the University of Cambridge, and Fellow of St John's College, has been Editor since 1956.

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What if you chose HP products to

For example: HP's new Oscilloscope Measurement System. The real surprise is not just that it's better, but that it is computer-programmable and extremely easy to use.



The HP 1980 simplifies performance of tests and adjustments on complex electronic equipment. **At left**, the display directs the operator to adjust a circuit under test. **Below right**, using programs stored in an external computer, the HP 1980 guides production tests with instructions to localize circuit defects quickly. **Below left**, the HP 1980 checks itself, quickly verifying its digital and analog operation.

The most obvious difference between the HP 1980 and conventional scopes is the front panel. The Hewlett-Packard system does away with the bewildering maze of knobs and switches scope users have had to master, and replaces them with a clean, logical array of color-coded keys and a single rotary control.

Internally, too, the HP 1980 is designed unlike any other oscilloscope. Controlled by a microprocessor, the HP 1980 reduces knob twiddling to the point that only minimal training is needed to operate it. (For instance, to obtain an initial display, only one key, AUTO-SCOPE, need be pressed.)

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freeing them for more creative, productive, and cost-effective work.

The HP 1980's automated operation—though both built-in and remote computer programming capabilities—brings new possibilities and time savings to product development and production test:

- HP-IB (IEEE-488) compatibility makes it possible for the HP 1980 to receive programming instructions from a computer, or to send measurement information to a computer for processing, evaluation, and hard copy printout.
- The simplified panel cuts setup time by as much as a factor of five. Front panel setups can be saved in the system's nonvolatile memory (they can't

be lost even if the system is turned off), then recalled and implemented with only two keystrokes.

• Calibration menus called up on the CRT display permit complete calibration via the front panel, with no operator's manual and minimal test equipment. What's more, the HP 1980 can be calibrated in its operating environment in less than 45 minutes—compared with the 6 to 10 hours it now takes trained technicians to calibrate conventional scopes in a calibration lab.

If you are developing or manufacturing products that use electronics technology, you owe it to yourself to find out more about the HP Oscilloscope Measurement System. Price of the HP 1980 is \$8500*.

enhance your skills?

For example: At Hoechst-Roussel, an HP 3000 computer system has helped cut the time it takes to complete a New Drug Application.

Hoechst-Roussel Pharmaceuticals, Inc., a subsidiary of American Hoechst Corporation in Somerville, New Jersey, develops and manufactures ethical drugs. At any given time, a dozen drugs and formulations are likely to be under clinical investigation.

Clinical testing of a new drug is a rigorous procedure; each step must have the approval of the Food and Drug Administration. To meet FDA requirements for a New Drug Application, the company must conduct studies that can involve thousands of patients across the country, who are given the drug under the supervision of their doctors. For each of these patients, a doctor typically completes from 10 to 250 documents describing data from various tests.

Faced with this massive amount of data to enter, validate, and reduce, Hoechst-Roussel's Group Director of Research Systems, Dr. Krishan Arora, found most data handling solutions unsatisfactory. "We needed a local computer with power and flexibility, one that could provide efficient data entry as well as on-line data base processing and inquiry," says Dr. Arora.

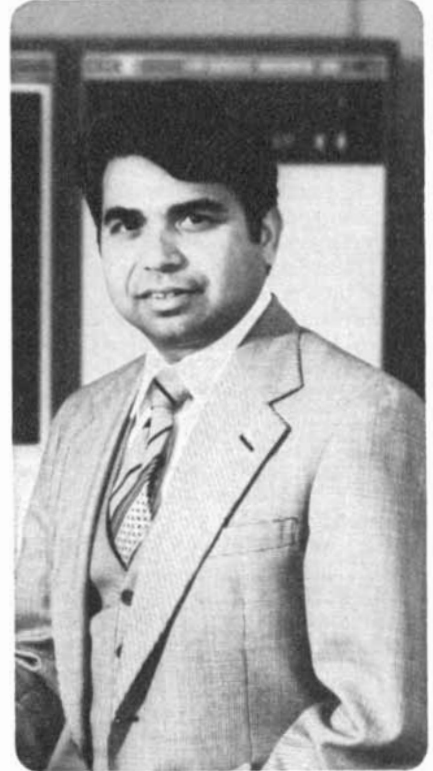
The search finally led to an HP 3000 Series III computer system. In conjunction with a large, mainframe computer, the HP 3000 gives the Medical Research Department a

powerful, easy-to-use tool for entering, editing, correcting, and reviewing the data that will later be analyzed on the mainframe.

"We wanted to save time, since data preparation—the data processing exclusive of the analysis—takes 60 to 75 percent of the time," says Dr. Arora. "The HP 3000 lets us organize our data in every relevant way. The system's flexible data base management gives us instant access to selected areas of information that we or the FDA might want: demography, critical side effects, or long range effects, for example."

Because the HP data base is interactive, the Medical Research Department can correct problems instantly. "We get data in the form we want, such as uniform listings or tables. We have been able to optimize new drug testing, streamline data processing procedures, and customize our output formats. Overall, we have doubled our output without increasing our staff. With the HP 3000, we have complete control over this very complex operation."

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the line is no longer one of the half-hour radial lines.

The entire pattern changes its appearance when the other correction is made, the one converting the sundial's local apparent time into its local mean time by means of the equation of time. Line 30 converts the equation of time into an angle: .25 degree for each minute of time. This angle and the longitude difference are displayed at the top of the computer screen. They are then combined and the total correction angle is shown. If the total is positive, the shadow points are to be rotated toward the west about the ecliptic. If the total is negative, the rotation is to be to the east.

Once the shadow points have been shifted according to the total correction angle the appearance of the pattern changes because the correction is different for each day of the year. (The daily difference comes from the equation of time, not from the longitude difference, which remains the same throughout the year.) Hence the shadow point for a particular time on a given day will shift in one direction from the half-hour line whereas the point for the same time on a different day will shift by a different amount or even in the other direction.

The result is both unexpected and pretty. If you draw a line through all the shadow points for any particular time of day, you see a distorted figure eight, an analemma. With this line you can tell the standard time on Sloan's sundial. Consider the analemma overlapping the noon line, which extends northward from the ecliptic. When the shadow tip passes the noon line, the local apparent time is noon. When the tip passes over the segment of the analemma corresponding to that day's date, the standard time (the time on a clock) is noon.

Each radial line from the ecliptic has an analemma. (If you make a small sundial grid, the pattern may be cluttered. Then you should draw an analemma around the radial lines corresponding to a full hour.) The local apparent time is read when the shadow passes a particular radial line, such as the one corresponding to 2:00 P.M. local time. When the shadow passes over the segment of the associated analemma correspond-

ing to that day's date, the standard time is 2:00.

To make your own analemmic sundial first change the values for the latitude and for the difference in longitude that are listed in line 10 of my computer program. (Use positive for west and negative for east and enter both in units of degrees.) Also replace the value for H to correspond to the length of your gnomon (leave the value in units of meters). Then run the program. As the computer requests, enter a date and the solar declination (in degrees) for that date. Also enter the equation of time (in minutes) for that date and whether the sundial is fast or slow compared with the local mean time. Tables of average values for the declination and equation of time can be found in the delightful book on sundials by Albert E. Waugh cited in the bibliography on page 246. More precise values can be obtained from the *American Ephemeris and Nautical Almanac*.

The program displays on the screen the difference in longitude (in degrees), the equation of time (converted into degrees) and the total angle by which the shadow points are to be shifted for that date. Then it displays the date and two columns of figures: one column has the time (in hours with respect to noon) and the other has the corresponding location of the shadow point on the north side (positive result) or the south side (negative result) of the equinoctial line.

To plot the grid lay out the ecliptic point and the equinoctial line. Mark the half-hour points along the equinoctial line and draw radial lines through them from the ecliptic point. The basic sundial grid of half-hour radial lines is then finished. Next Sloan's modifications are to be made.

The distance between the radial lines represents half an hour, or 7.5 degrees of (apparent) rotation of the sun about the earth. Along the equinoctial line between the radial lines mark off units of one degree. Each unit will have a length equal to the tangent of one degree multiplied by the secant of the sundial's latitude and by the height of the post. These additional marks will help you to make the corrections of the shadow-point locations.

Run the program to cover 34 days of the year. Each run will yield a set of shadow points to plot on the grid, one point for each of the half-hour radial lines. Suppose that for the radial line corresponding to two hours after noon the distance is .2 meter on the north side of the equinoctial line and the total angle is two degrees to the west. The shadow point should be plotted on a line through the ecliptic point that is rotated two degrees to the west from the 2:00 radial line. Use the one-degree marks on the equinoctial line to find the correct place. On the rotated line (which is not drawn) the shadow point is .2 meter on the north side of the equinoctial line. Label the point with the date. All the shadow points plotted for that date will be shifted from the main radial lines by the same angle, so that the plotting goes quite fast.

After you have plotted the points for the 34 days draw an analemma for each of the half-hour radial lines. The grid for Sloan's analemmic sundial is then complete. The local apparent time is read from the half-hour radial lines, as would be done with a standard sundial. The standard time is read from the analemma that overlaps each radial line. If daylight-saving time is in effect, change the standard time by an hour.

The computations could be done with a pocket calculator, but they would take much more time. Readers skilled in a language for a home computer will be able to improve on my program. If analemmic sundials are to be designed for many locations, the values for declination and equation of time should be stored in the computer to eliminate the need for responding with the data when the computer needs it.

The appearance of the pattern varies with latitude in an interesting way. For places progressively closer to the Equator the half-hour radial lines become shorter because the sun is typically higher in the sky. The analemma overlapping each of the lines becomes a more perfect figure eight perpendicular to the equinoctial line. For the North Pole the pattern would be very different. (The pattern would of course be useful only for the spring and summer months, since the sun is not visible for the rest of the year.) The pattern is circular, with the noon radial line on one side and the corresponding line for midnight on the other side. The analemma consists of single loops instead of two.

Sundials similar to Sloan's were probably built long ago by the patient plotting of shadows rather than by calculations. I think it would be fun to mimic the ancients by building a huge sundial, rivaling in size the standing stones in Britain and France. Presumably the grid could be made accurate enough to incorporate the yearly changes in the declination of the sun and in the equation of time.

```

10 CLS : RPD = .01745 : L= 35.0 * RPD : DL = 1.8 : H = 1.0
20 INPUT "DECLINATION (DEG) =" : DECL : INPUT "DATE =" : A$
30 INPUT "EQUATION OF TIME (MIN) =" : ET : ET = ET * 0.25
40 INPUT "IS DIAL FAST OR SLOW" : B$ : D = DECL * RPD
50 IF B$ = "FAST" THEN ET = -ET
60 PRINT "DL =" : DL, "EQ TIME =" : ET, "TOTAL ANGLE =" : DL + ET
70 PRINT " " : PRINT "TIME (HRS)", "DIST (METERS)", A$
80 X = TAN (L) : F = 1/ COS (L) : G = X + 1/X : W = 1/ SIN (L)
90 FOR K = 0 TO 10 : C = K * 7.5 * RPD : B = TAN (C)
100 E = SQR ( (B * F) ^ 2 + G ^ 2) / G : J = 1/ (E * X)
110 A = ATN ( (G * E - J) / SQR (W ^ 2 - J ^ 2) )
120 Z = TAN (A - D) / TAN (A) : DIST = H * (Z - 1) * (G * E - J)
130 PRINT T, DIST : T = T + .5 : NEXT K : END

```

A computer program to calculate shadow-tip positions for a sundial in the Northern Hemisphere